7 **(a)** State what is meant by a photon.

.....

.....

..... [2]

(b) Electromagnetic radiation of a varying frequency f and constant intensity I is used to illuminate a metal surface. At certain frequencies, electrons are emitted from the surface of the metal. The variation with f of the maximum kinetic energy E_{MAX} of the emitted electrons is shown in Fig. 7.1.

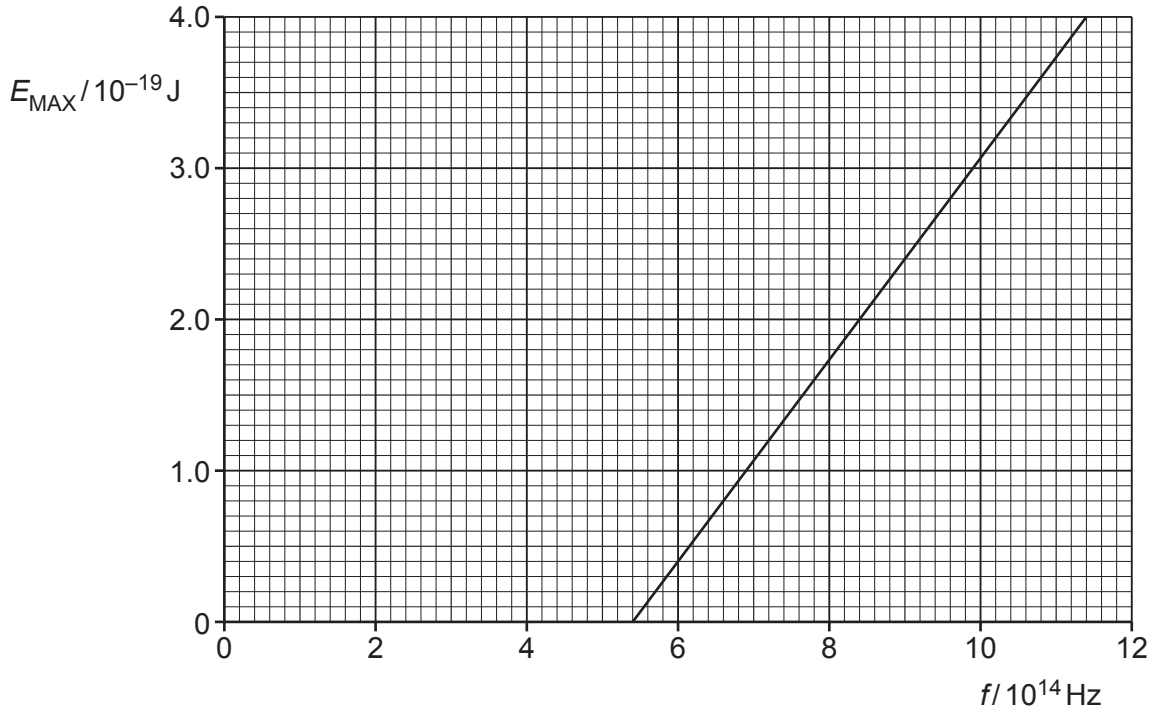


Fig. 7.1

(i) State the name of this phenomenon.

..... [1]

(ii) Describe **three** conclusions that can be drawn from the graph in Fig. 7.1. The conclusions may be qualitative or quantitative.

1 .....

.....

2 .....

.....

3 .....

.....

[3]

(c) The experiment in **(b)** is repeated twice, each time making one change.

State, with a reason, how the graph obtained would compare with Fig. 7.1 when:

(i) a different metal is used, but keeping the intensity I of the radiation the same

.....

.....

..... [2]

(ii) the same metal is used, but with electromagnetic radiation of intensity $2I$.

.....

.....

..... [2]

[Total: 10]